Theorem: Let F/K be a radical extension and let $E \subseteq F$ be Galois over K. Then $\mathrm{Gal}(E/K)$ is solvable.

Proof: Without loss of generality, say F is Galois over K. Then $\mathrm{Gal}(E/K) \simeq \mathrm{Gal}(F/K)/\mathrm{Gal}(F/E)$ and homomorphic images of solvable groups are solvable, so it is enough to show that $\mathrm{Gal}(F/K)$ is solvable.

Indeed, let $F = K(\alpha_1, \dots, \alpha_n)$ where $\alpha_1^{m_1} \in K$ for some $m_1 \in \mathbb{N}$, $\alpha_2^{m_2} \in K(\alpha_1)$ for some $m_2 \in \mathbb{N}$, $\alpha_3^{m_3} \in K(\alpha_1, \alpha_2)$ for some $m_3 \in \mathbb{N}$, etc...

Let ζ be a primitive $m = (m_1 \cdots m_r)$th root of unity.

$$\begin{array}{ccc} & F(\zeta) & \\ / & & \backslash \\ F & & K(\zeta) \\ \backslash & & / \\ & K & \end{array}$$

We know that $F(\zeta)$ is an abelian extension of F and $K(\zeta)$ is an abelian extension of K.

Claim: To show that $\mathrm{Gal}(F/K)$ is solvable, it suffices to show $\mathrm{Gal}(F(\zeta)/K(\zeta))$ is solvable.

Note that $\mathrm{Gal}(F(\zeta)/F) \triangleleft \mathrm{Gal}(F(\zeta)/K)$, so then $\mathrm{Gal}(F/K) \simeq \mathrm{Gal}(F(\zeta)/K)/\mathrm{Gal}(F(\zeta)/F)$, so it suffices to show $\mathrm{Gal}(F(\zeta)/K)$ is solvable.

Now, $K(\zeta)$ is Galois over K, so we know that $\mathrm{Gal}(F(\zeta)/K(\zeta)) \triangleleft \mathrm{Gal}(F(\zeta)/K)$ and also the quotient $\mathrm{Gal}(F(\zeta)/K)/\mathrm{Gal}(F(\zeta)/K(\alpha)) \simeq \mathrm{Gal}(K(\zeta)/K)$, so

Lemma: If $N \triangleleft G$ and N and G/N are solvable, then

G is solvable.

So $F(\zeta)/K(\zeta)$ is solvable, so F/K is solvable.

In particular, $F(\zeta) = K(\alpha_1, \dots, \alpha_r, \zeta)$. Let

$G_{r-i} = Gal(F(\zeta)/K(\alpha_1, \dots, \alpha_i)) \leq Gal(F(\zeta)/K(\zeta))$

$G_r = \{e\}$

$G_r \leq G_{r-1} \leq G_{r-2} \leq \cdots \leq G_0 = Gal(F(\zeta)/K(\zeta))$

Then $G_{j+1} = Gal(F(\zeta)/K(\zeta, \alpha_1, \dots, \alpha_{j+1})) \triangleleft Gal(F(\zeta)/K(\zeta, \alpha_1, \dots, \alpha_j))$

since $K(\alpha_1, \dots, \alpha_{j+1})$ is a Galois extension of $K(\zeta, \alpha_1, \dots, \alpha_j)$

because $\alpha_{j+1}^{m_{j+1}} \in K(\zeta, \alpha_1, \dots, \alpha_j)$ and $K(\zeta, \alpha_1, \dots, \alpha_j)$ contains

a primitive m_{j+1}th root of unity $\zeta^{m_1 \cdots m_j / m_{j+1}}$ so then

$G_j/G_{j+1} = Gal(F(\zeta)/K(\zeta, \alpha_1, \dots, \alpha_j))/Gal(F(\zeta)/K(\zeta, \alpha_1, \dots, \alpha_{j+1}))$

$\simeq Gal(K(\zeta, \alpha_1, \dots, \alpha_{j+1})/K(\zeta, \alpha_1, \dots, \alpha_j))$

$= Gal(K_j(\alpha_{j+1})/K_j)$ (abelian)

so $Gal(F(\zeta)/K(\zeta))$ is solvable, so $Gal(F/K)$ is solvable.

so $Gal(E/K)$ is solvable.

Transcendental Extensions

Recall: If F is an extension over K, $\alpha \in F$ is algebraic over K if $f(\alpha) = 0$ for some nonzero $f \in K[x]$ and transcendental otherwise.

Definition: $\alpha_1, \dots, \alpha_r \in F$ are algebraically dependent over K if there is a nonzero polynomial $f \in K[x_1, \dots, x_r]$ such that $f(\alpha_1, \dots, \alpha_r) = 0$, and algebraically independent otherwise.

A subset of F is algebraically dependent if every finite subset is.

Example: π and π^2 are transcendental but not algebraically dependent.

Theorem: $\alpha_1, \dots, \alpha_r \in F$ are algebraically independent over K if and only if $K(\alpha_1, \dots, \alpha_r) \simeq K(x_1, \dots, x_r)$

Proof: There is an epimorphism $K[x_1, \dots, x_r] \twoheadrightarrow K[\alpha_1, \dots, \alpha_r]$ defined by $f \mapsto f(\alpha_1, \dots, \alpha_r)$ and algebraically independent if and only if $\ker = \{0\}$. So if $\alpha_1, \dots, \alpha_r$ are algebraically independent, this is an isomorphism $K[\alpha_1, \dots, \alpha_r] \simeq K[x_1, \dots, x_r]$ so $K(\alpha_1, \dots, \alpha_r) \simeq K(x_1, \dots, x_r)$.

Definition: Let F/K be an extension. A transcendental basis for F/K is an algebraically independent set $S \subseteq F$ that is maximal.

Zorn's lemma always gives a transcendental basis.

Example: $\emptyset$ obvious for any algebraic extension. $\{x\}$ is a transcendence basis for $K(x)$ over K. In particular, x is transcendental, so $\{x\}$ is algebraically independent. If $f(x) \in K[x]$ such that $f(x) \neq x$, if $f(x) \in K$, then $\{f(x)\}$ is algebraically dependent, so $\{x, f(x)\}$ is algebraically dependent. On the other hand, if f is non constant, then $(y_1, y_2) = (x, f(x))$ is a root of $y_2 \, \mathrm{denom}(f)(y_1) - \mathrm{numer}(f)(y_1)$, so $\{f(x)\}$ is a transcendence basis iff $f(x) \notin K$

Note: If $S \subseteq F$ is a TB for F/K, then F is algebraic over $K(S)$.

Theorem: Let F/K be an extension with $S \subseteq F$ algebraically independent. Then $S \cup \{\alpha\}$ is algebraically dependent if and only if α is transcendental over F/K

Proof: If S is algebraically independent, and $S \cup \{\alpha\}$ is algebraically dependent. Then there exists $s_1, \dots, s_r \in S \cup \{\alpha\}$ nonzero $f(x_1, \dots, x_r) \in K[x_1, \dots, x_r]$ such that $f(s_1, \dots, s_r) = 0$. Since S is algebraically independent, $\alpha \in \{s_1, \dots, s_r\}$. Without loss of generality, say $s_r = \alpha$.

$f(s_1, \dots, s_{r-1}, y) \in K(s_1, \dots, s_{r-1})[y]$ is a nonzero polynomial which vanish at α. So α is algebraic, $K(s_1, \dots, s_{r-1}) \subseteq K(S)$.

Conversely, if α is algebraic over $K(S)$, then $f(\alpha) = 0$ for some $f(x) \in K(S)[x]$. On finitely many elements of S show up in coefficients, so $f(x) \in K(s_1, \dots, s_r)[x]$ for some $s_1, \dots, s_r \in S$.

Because $K[s_1, \dots, s_r] \cong K[x_1, \dots, x_r]$ so $f(x) = g(s_1, \dots, s_r, x)$

where $g(x_1, \dots, x_r, x) \in K[x_1, \dots, x_r, x]$,

$$g(s_1, \dots, s_r, \alpha) = 0 \Rightarrow \{s_1, \dots, s_r, \alpha\} \text{ alg. dep.}$$

$$\Rightarrow S \cup \{\alpha\} \text{ alg. dep.}$$

Corollary: Let F/K be a field extension. Then an algebraically independent set $S \subseteq F$ is a transcendence basis if and only if $F/K(S)$ is algebraic.

Example: $\mathbb{C}(x, \sqrt{x^3+1})$ is an extension of $\mathbb{C}$ and $\{x\}$ is a transcendence basis as $\mathbb{C}(x, \sqrt{x^3+1})$ is algebraic over $\mathbb{C}(x)$, but $\mathbb{C}(x, \sqrt{x^3+1}) \neq \mathbb{C}(x)$.

Theorem: Let F/K be a field extension. Then every transcendence basis for F/K has the same cardinality.

Definition: The transcendence rank of F/K is the cardinality of any transcendence basis, $\mathrm{TrD}(F/K)$.

Theorem: $\mathrm{tr.d}(F/K) = \mathrm{tr.d}(F/E) + \mathrm{tr.d}(E/K)$.